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In-depth single-cell transcriptomic exploration of the regenerative dynamics in stony coral

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Coral reef ecosystems face escalating threats from anthropogenic global climate challenges, leading to frequent bleaching events. A key issue in coral transplantation is the inability of fragments to rapidly grow to sizes that can resist environmental pressures. The observation of accelerated growth during the early stages of coral regeneration provides new insights for addressing this challenge. To investigate the underlying molecular mechanisms, we study the fast-growing stony coral *Acropora muricata*. Using single-cell RNA sequencing, bulk RNA sequencing, and high-resolution micro-computed tomography, we identify a critical regeneration phase around 2–4 weeks post-injury. Single-cell transcriptome analysis reveals 11 function-specific cell clusters. Pseudotime analysis indicates epidermal cell differentiation into calicoblasts. Bulk RNA-seq results highlight a temporal limitation in coral's rapid regeneration. Through integrated multi-omics analysis, this study emphasizes the importance of a comprehensive understanding of coral regeneration, providing insights beyond fundamental knowledge and offering potential protective strategies to promote coral growth.

Coral reef ecosystems are undergoing a transformative phase with increasingly brief intervals between coral bleaching events, impeding the recovery of mature coral colonies. As of the 1880s, Earth's surface temperature has surged by nearly 1 °C due to anthropogenic climate warming and El Niño, continuously setting unprecedented records^{1,2}. Recent global sea surface temperature spikes further accelerate the warming trend, reducing the median return time between severe coral bleaching events to just 6 years by 2016^{3,4}. This escalating frequency poses a significant threat to coral reefs, vital thermal refuges for diverse marine species⁵. Ominous predictions suggest many reefs may vanish by midcentury⁴.

Faced with this challenge, urgent and informed management strategies are crucial to protect coral reefs and restore their compromised structure and ecological functionality. Globally, researchers, conservationists, and environmental managers are actively devising innovative strategies to preserve coral reef ecosystems amidst diverse local and global threats⁶. Genetic and reproductive interventions, like managed selection and breeding, aim to propagate stress-tolerant corals⁷. Techniques such as gamete and larval capture⁸, coral cryopreservation⁹, and genetic manipulation¹⁰ offer novel

ways to enhance genetic diversity. Physiological interventions, including pre-exposure¹¹ and algal symbiont manipulation¹², tap into corals' adaptive abilities. Strategies like microbiome manipulation¹³, antibiotic application¹⁴, and phage therapy¹⁵ target coral microbial communities for improved health and stress resistance. Coral population interventions, like managed relocation¹⁶, propose shifting corals to favorable environments. Environmental strategies, including shading¹⁷ and seawater carbonate chemistry¹⁸ interventions, suggest innovative ways to mitigate thermal stress and modify seawater chemistry. Despite implemented measures, coral reefs decline unabated, with a 14% loss of coral cover globally between 2009 and 2018, which exceeds the total coral currently found on Australia's coral reefs¹⁹. One of the crucial factors is the need for substantial recovery volume to preserve reef structure while mitigating coral losses²⁰, based on the premise that growth should not come at the expense of genetic diversity and ecosystem stability. Recognizing this, micro-fragmentation leverages corals' regenerative capacity, segmenting colonies for transplantation. Research indicates accelerated growth in micro-fragments post-fragmentation^{21,22}, highlighting its potential efficacy. However, molecular mechanisms remain

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unclear. Lock et al.²³ proposes that micro-fragmentation in *Porites lobata* disrupts calcium homeostasis, energizing coral expansion as a wound healing response, contributing to survival. The study hypothesizes enhanced calcification rates for skeletal regeneration, but understanding these molecular processes is limited.

To uncover the molecular mechanisms driving stony coral regeneration, we focus on *Acropora muricata*, a fast-growing, perforate coral crucial for reef structure²⁴. As a perforate coral, the tissue and skeleton of *A. muricata* are highly interconnected, which makes it particularly well-suited for studying coral regeneration. Our study integrated scRNA-seq, bulk RNA-seq, full-length transcriptome sequencing, and miRNA analysis for a holistic view of coral regeneration. Differences between scRNA-seq and bulk RNA-seq data emphasize the complex regulatory processes. The multifaceted approach aimed to elucidate cellular dynamics, gene expression, and communication networks, contributing to coral regeneration understanding.

Results

Morphological and high-resolution micro-computed tomography (HRCT) 3-dimensional (3D) reconstruction of coral regeneration

In our exploration of optimal regeneration stages for *A. muricata*, previous laboratory observations indicated that the peak regeneration period occurs

approximately 2–4 weeks post-injury. Building on this knowledge, we systematically collected samples from days 0 to 39, with observations every 3 days. Skeletal morphological studies revealed that the most pronounced recovery occurred between 15 and 27 days (Fig. 1a). To validate these findings, selected samples from normal and days 14, 21, and 28 post-injury underwent HRCT 3D reconstructions (Fig. 1b). The cross-section of the normal coral sample clearly exhibits a radial symmetry structure. After 14 days of injury, the columella cross-section becomes irregular, indicating calcified substance deposition. By day 21, the deposition increased, completely filling the columella. In the longitudinal section, there is an obvious increase in calcified substance deposition near the injury site, indicating further biomineralization. By day 28, the cross-section again displays a radial symmetry structure, although it is not as pronounced as in the normal sample, signifying the late regeneration stage. Therefore, the regenerative capacity of coral peaks around day 21, prompting the selection of samples from this time point for subsequent single-cell level bioinformatics analysis, comparing them with normally growing coral samples.

Establishment of the *A. muricata* cell atlas utilizing 10x Genomics

We analyzed 9,053 filtered cells from scRNA-seq datasets of healthy ($n = 4,941$) and regenerating ($n = 4,112$) *A. muricata* coral. Using established cell markers (detailed in Supplementary Data 1), we identified 13 cell

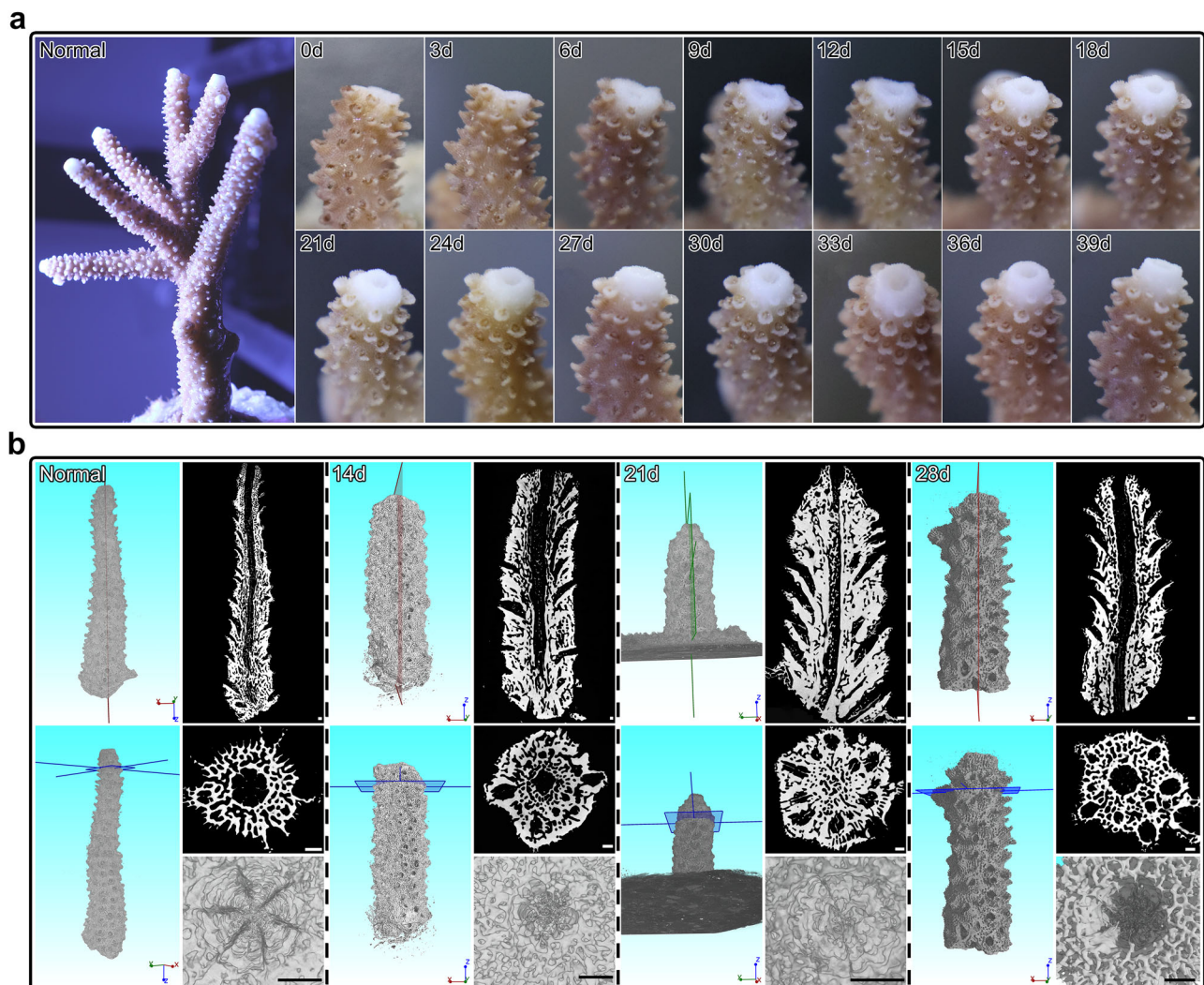


Fig. 1 | Monitoring coral regeneration growth and HRCT 3D reconstruction. **a** Left: Normal *Acropora muricata* branch before wounding; Right: Growth monitoring every 3 days post-injury. **b** 3D reconstruction of coral regeneration across four stages using HRCT. Each stage consists of two parts: Top left: Position of the

longitudinal section; Top right: Longitudinal cross-section of the coral skeleton. Bottom left: Position of the transverse section; Bottom right: Transverse cross-section of the coral skeleton, along with a 3D reconstruction showing a transverse cross-sectional view of the columella. Scale bar: 0.5 mm.

types, including gastrodermal cells, neurons, calicoblasts, symbiocytes (alga-hosting cells), epidermal cells, cnidocytes, gland cells, progenitor cells, immune cells, muscle cells, digestive filament cells, and two unassigned cell clusters (Supplementary Fig. 1). These unassigned clusters lacked cell-type-specific markers (Supplementary Data 1), and were therefore excluded from downstream functional analyses to avoid misinterpretation of ambiguous signals. Subsequent re-clustering of the 11 function-specific cell clusters was conducted for further analysis (Fig. 2a, b; Supplementary Figs. 2–8; Supplementary Data 1).

To validate cell clustering reliability, we used UCell to score Seurat objects based on cell markers and visualized cell type composition within clusters (Fig. 2c). Most clusters were predominantly composed of cells from their respective types, except for cnidocytes and digestive filament cells, which contained abundant immune cells. To compare cell distribution between treatments, we generated a t-SNE plot (Fig. 2d), with “Unwounded” representing the unwounded treatment and “Wounded” representing the healing treatment with peak biomineralization. Both treatments exhibited the 11 identified cell types. Despite total cell number variations, the relative proportions and distribution of most cell types within and across both treatments remained consistent (Fig. 2e, f), with the exception of calicoblasts and epidermal cells, which showed notable differences in cell proportions between treatments. Specifically, calicoblasts constituted 19.54% of cells in the unwounded sample compared to 5.32% in the wounded sample, while epidermal cells increased from 6.03% in the unwounded to 17.73% in the wounded sample.

To identify additional cell-type-specific markers, we analyzed highly expressed genes conserved across conditions. Notably, certain genes, distinct from those previously mentioned, exhibited consistently higher expression within the same cell types across both treatments (Supplementary Fig. 9 and Supplementary Data 2). For instance, *sulfate transporter* and *saccharopine dehydrogenase* showed specific high expression in symbiocytes. Additionally, *carbonic anhydrases 2* (*Ca2*), which is frequently reported as a key gene in biomineralization, also showed heightened expression in symbiocytes. Leucine-rich repeat extensin-like proteins showed high expression in cnidocytes. Immune cells displayed high expression of RIG-I-like receptor, an antiviral protein linked to the innate immune response²⁵. Consistent with known neural functions, genes associated with calcium ion binding (GO:0005509), encompassing calretinin, neurocalcin homolog, and calmodulin, exhibited high expression within neurons, aligning with their expected roles in neural signal modulation and homeostasis.

Exploring differential gene expression patterns across conditions

Utilizing the Wilcoxon rank-sum test, we compared single-cell transcriptomic data between wounded and unwounded coral samples, identifying 536 significantly differentially expressed genes (DEGs) (Fig. 3 and Supplementary Data 3). These DEGs were distributed across various cell types, with notable variations in the number of up- and downregulated genes within each cell type (Fig. 3a). Immune and muscle cells did not show significant DEGs in this analysis. A majority of DEGs were observed in a single cell type, with the exception of 9 and 79 genes, up- and downregulated, respectively, found in two or more cell types.

Further analysis of up- and downregulated DEGs from the wounded treatment compared to unwounded revealed significant enrichment in 13 and 21 Gene Ontology (GO) pathways, respectively (Fig. 3b and Supplementary Data 4). Among the upregulated DEGs, the GO pathway associated with the extracellular space (GO:0005615) exhibited the highest number of enriched genes (Fig. 3d). Notably, *integral membrane protein 2B* (*Itm2b*), which was also associated with the regulation of cell population proliferation (GO:0042127), exhibited remarkable upregulation in seven distinct cell clusters, including symbiocytes, calicoblasts, cnidocytes, epidermal cells, gastrodermal cells, gland cells, and neurons (Fig. 3c, d). Immune-related genes like *procathepsin L* (*Ctsl*), *prosaposin* (*Psap*, which may play a role in preventing oxidative stress²⁶), and pregnancy zone proteins (PZPs,

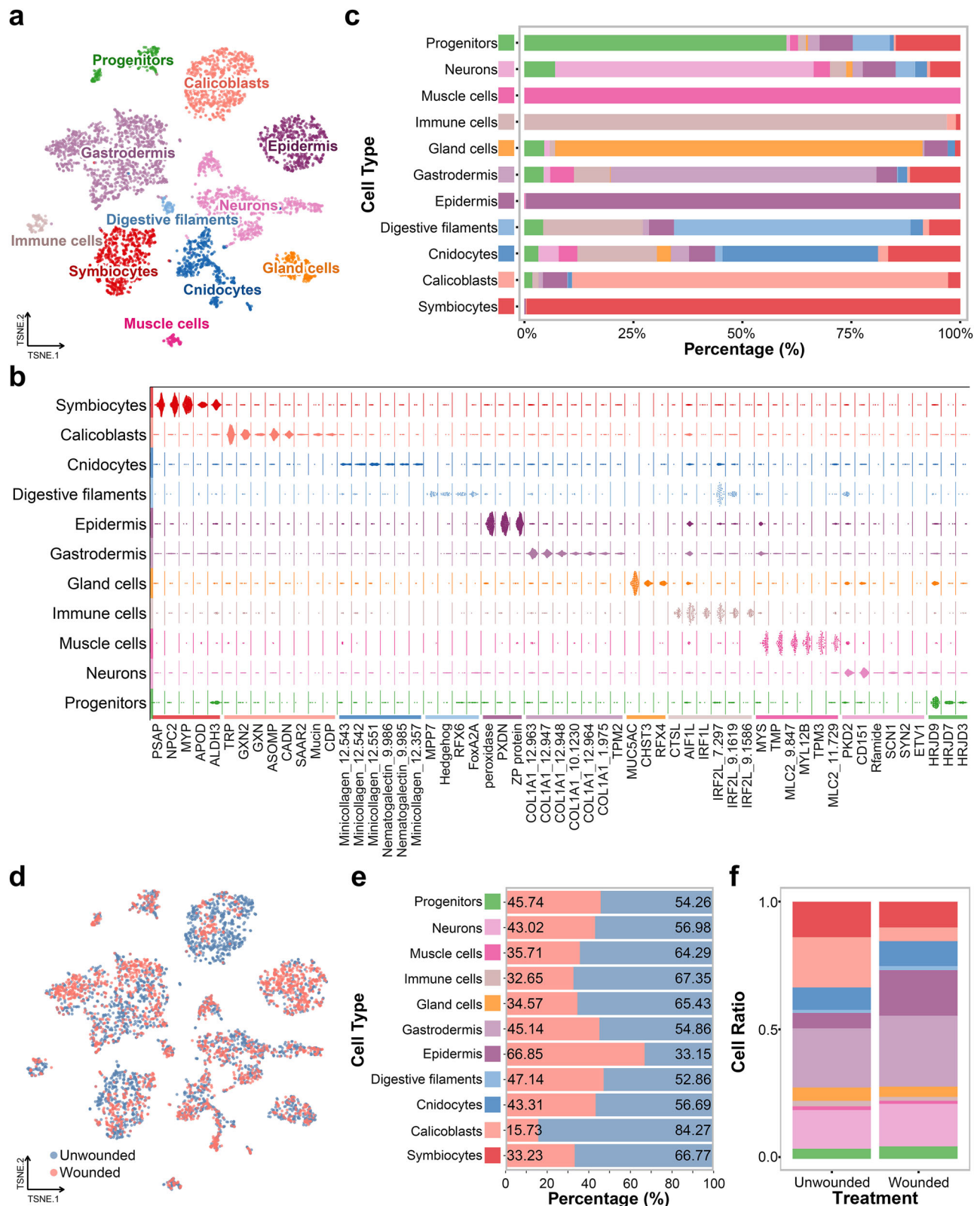
identified in the skeletal proteome of the sea star *Patiria miniata*²⁷) were also upregulated and enriched in the extracellular space pathway, specifically in the wounded treatment (Fig. 3b, d). Additionally, epidermal cells showed enrichment in the heme binding pathway (GO:0020037), including genes like *allene oxide synthase-lipoxygenase* (*AOS-LOX*), catalase (*Cat*), cytochrome b-245 light chain (*Cyba*), and *peroxidase homolog* (*Pxdn*) (Fig. 3b–d). These genes are classified as oxidoreductases and play a critical role in the coral’s response to oxidative stress, showing notable upregulation in the wounded treatment.

DEGs related to collagen trimer (GO:0005581) and extracellular matrix (ECM) structural constituent (GO:0005201), which were upregulated in the wounded treatment, were predominantly concentrated in the gastrodermis (Fig. 3b–d). These genes belong to the alpha collagen group, a key component of the extracellular matrix critical for regulating wound healing phases²⁸. In calicoblasts and symbiocytes, there was significant upregulation of *Ca2* associated with carbonate dehydratase activity (GO:0004089) (Fig. 3b–d). Genes involved in iron ion transport (GO:0006826), such as ferritin in gland cells and transferrin in symbiocytes, were upregulated in the wounded treatment (Fig. 3b–d). Ferritin plays a crucial role in maintaining iron homeostasis and protecting against oxidative damage caused by free iron and ROS²⁹. Transferrin, a conserved iron transport protein in metazoans, facilitates the delivery of iron to symbiotic algae through transferrin-positive vesicles³⁰. Additionally, DEGs enriched in lipid droplet (GO:0005811) and upregulated in the wounded treatment were identified in symbiocytes, including *long-chain-fatty-acid-CoA ligase 4* (*Acsf4*), which is involved in fatty acid metabolism, and *saccharopine dehydrogenase-like oxidoreductase* (*Scdpdh*) (Fig. 3b–d).

For the downregulated DEGs identified in the wounded treatment, a group primarily located in the gastrodermis exhibited enrichment in multiple GO terms, particularly in ribosomal proteins and elongation factors, essential components of ribosomes and cyclically associated with ribosomes during protein synthesis elongation (Supplementary Fig. 10). Another group of these DEGs related to DNA-binding transcription factor activity, RNA polymerase II-specific (GO:0000981), was prominently expressed in the epidermal cells, calicoblasts, and gastrodermal cells (Fig. 3c, e). DEGs involved in proton motive force-driven ATP synthesis (GO:0015986), primarily composed of components of the cytochrome c oxidase complex, displayed downregulation in the gastrodermis of the wounded treatment (Fig. 3c, e). Cytochrome c oxidase is a crucial enzyme complex responsible for the final step of the electron transport chain in mitochondria, a process essential for generating cellular energy in the form of ATP^{31,32}. Additionally, a group of DEGs related to protein folding, including *heat shock protein HSP 90-alpha* (*Hsp90aa1*) and *DnaJ protein homolog 1* (*Dnajb1*), showed significant downregulation in the gastrodermal cells, calicoblasts, epidermal cells, and gland cells in the wounded treatment (Fig. 3c, e). Finally, a set of protein phosphatase 1 regulatory (PPP1R) subunits derived from the gastrodermal cells and calicoblasts were enriched in GO:0008287, related to the protein serine/threonine phosphatase complex, with downregulation in the wounded treatment (Fig. 3c, e).

Changes in cell-cell communication networks during coral regeneration

To gain deeper insights into the intricate cell-cell communication networks orchestrating the regeneration process, we conducted a comparative analysis using CellChat software. The results unveiled significant associations of intercellular communication in both treatments with the ECM–receptor regulatory network, implicating key ECM proteins such as collagen, thrombospondin (THBS), heparan sulfate proteoglycan (HSPG), and laminin, play pivotal roles in modulating cell–cell and cell–matrix interactions^{33–36} (Supplementary Data 4). Noteworthy differences in interactions among these pathways emerged during coral regeneration, with increased interactions directed towards digestive filaments and muscle cells, and decreased interactions targeting calicoblasts, epidermal cells, and gastrodermal cells (Fig. 4a). The interaction strength between gastrodermal cells and calicoblasts showed a pronounced decline. Mapping the inferred



intercellular communication networks onto a shared two-dimensional manifold revealed clustering based on functional similarity (Fig. 4b). Additionally, the overall communication probability between the two treatments showed that THBS and laminin were highly active in unwounded samples, with respective contribution values of 0.13 and 0.10 in the unwounded group, compared to 0.10 and 0.08 in the wounded group (Fig. 4c).

In-depth analysis of outgoing and incoming interaction strength pinpointed calicoblasts as the cell population exhibiting the most significant changes in sending or receiving signals during coral regeneration (Fig. 4d). Building upon this, we investigated the signaling ligand-receptor pairs directed from calicoblasts towards other cell types (Fig. 4e, f, and Supplementary Data 4). The results revealed heightened and upregulated LAMC1-

Fig. 2 | *Acropora muricata* cell atlas of regeneration. **a** The integrated transcriptomes of the wound and unwound treatments of *A. muricata*, obtained through scRNA-seq, were grouped into 11 clusters (color-coded) and visualized in t-SNE space. Each colored dot represents an individual cell. **b** Beeswarm plots showing the log-transformed UMI counts of marker genes (on the x-axis) for each cell type (on the y-axis). Different colors represent distinct cell types, consistent with (a). **c** Stacked bar plot showing the percentage of cells from each cell type (y-axis) being classified as different cell types (x-axis), based on UCell scoring. Each bar represents the distribution of classification scores for the cells within a specific cell type, illustrating potential overlaps and the assessment of one cell type as another. **d** t-SNE visualization of the integrated transcriptomes from the wound and unwound treatments of *A. muricata*, color-coded by treatment group. Each dot represents an

individual cell, and the plot shows the distribution of cells across 11 clusters according to their respective treatments, highlighting differences in cellular composition between wound and unwound conditions. **e** Stacked bar chart showing the distribution of cell types across wound and unwound treatments. The y-axis represents the different cell types, and the x-axis shows the percentage of cells within each treatment group. Each bar is color-coded to represent the proportion of cells in each treatment (wound vs. unwound), with the corresponding percentage values displayed on the bars (%). **f** Stacked bar chart showing the proportion of each cell type in the wound and unwound treatments. The y-axis represents the proportion of each cell type, and the x-axis displays the two treatment conditions (wound vs. unwound). Each color represents a different cell type, with the relative proportion of each cell type in the total population of cells for each treatment shown in the chart.

DAG1 signaling from calicoblasts to digestive filaments and muscle cells (Supplementary Fig. 11a). Additionally, a surge in THBS2-SDC4 signaling was observed towards immune cells, while THBS2-SDC1 signaling towards multiple cell types diminished (Supplementary Fig. 11b). Conversely, signaling ligand-receptor pairs targeting calicoblasts from other cell types showed no enhancement in signals but identified weakened and down-regulated COL1A2-SDC1 signaling from gastrodermal cells to calicoblasts (Supplementary Figs. 12, 13, and Supplementary Data 4).

Differentiation of epidermis during coral regeneration

In stony coral development, certain epidermal cells transform into calicoblasts, initiating rapid skeletal deposition^{37,38}. Pseudotime analysis was employed to explore the relationship and potential transitions between these cell types during coral regeneration (Fig. 5). Results showed that, during coral development, some epidermal cells transitioned into two distinct calicoblast types (Fig. 5a–c). In contrast, during regeneration, more epidermal cells predominantly transformed into a single subtype of calicoblasts (Fig. 5d). We constructed a heatmap and performed GO enrichment analysis to explore the dynamic expression changes of genes associated with cell transformation (Fig. 5e and Supplementary Data 5). Under unwounded treatment, epidermal cells exhibit three distinct cell density peaks, which may indicate different developmental stages. This observation aligns with the pseudotime analysis results, where three stages of epidermal cell development are identified; however, these stages are positioned along a single branch in the trajectory, suggesting no distinct branching into separate developmental paths (Supplementary Fig. 14). Similarly, calicoblasts under unwounded treatment display three density peaks from the initial state to maturity, but notably with a relatively higher cell proportion at the mature stage. During regeneration, however, both epidermal cells and calicoblasts in earlier developmental stages show a significant increase in numbers.

In the early phases of calicoblast development during regeneration, specifically in cluster 1 and cluster 3, several genes exhibited distinct upregulation, prominently enriched within the extracellular region (GO:0005576), with proteins like threonine-rich protein, aspartic and glutamic acid-rich protein, galaxin, acidic skeletal organic matrix protein, CUB domain-containing protein (*Cdcp1*), and skeletal aspartic acid-rich protein 1 (*Saarp1*) (Fig. 5e, f and Supplementary Data 5). These proteins are known components of the coral skeleton^{39–41}. Additionally, we identified AOS-LOX, peroxinectin, and melanotransferrin, which are associated with immune response^{42,43}, showing similar expression patterns (Supplementary Fig. 15a and Supplementary Data 5). Furthermore, cluster 3 showed specific upregulation of *galaxin 2*, *Saarp2*, and *Mucin* (Supplementary Fig. 15b and Supplementary Data 5). In the advanced developmental phases of calicoblasts within cluster 4, limited genes, including *uncharacterized skeletal organic matrix protein 4* (*Usomp4*), *Matrix metalloproteinase 25* (*Mmp25*), and *metalloproteinase inhibitor 1* (*Timp1*), exhibited distinct high expression (Supplementary Fig. 15c and Supplementary Data 5).

The transformation of coral epidermal cells into calicoblasts resulted in the emergence of two distinct fates. To better understand the differences between these fates, we conducted a GO enrichment analysis (Fig. 5g and Supplementary Data 5). Genes associated with cell fate 1 (state 1 shown in

Supplementary Fig. 14), primarily composed of calicoblasts in the unwounded treatment, were significantly upregulated in two clusters (1 and 3). Cluster 1 exhibited enrichment in extracellular matrix and metalloendopeptidase activity, including genes such as *Mmp24*, *Mmp25*, *blastula protease 10* (*Bp10*), and *Timp1* (Fig. 5h). Cluster 3 showed enrichment in processes related to protein stabilization and apoptotic signaling pathways, involving genes like *Hsp90aa1*, *Dnajb1*, *Bcl2*, and *apoptosis regulator 1* (*Ar1*) (Supplementary Fig. 16). Genes associated with cell fate 2 (state 5 shown in Supplementary Fig. 14), which includes the majority of calicoblasts from the wounded treatment and a smaller subset from the unwounded treatment, were enriched in proton motive force-driven ATP synthesis, transmembrane transporter complex, and regulation of cell population proliferation (Supplementary Fig. 17). Additionally, we identified a set of upregulated genes associated with biomineralization in cell fate 2, including *Ca2*, *Saarp2*, *galaxin*, *galaxin2*, and *Mucin*. In cell fate 1, we observed the upregulation of *Usomp4*, *Cdcp2*, and *calumenin* (Supplementary Fig. 18). These results highlight differential gene expression during the transition of epidermal cell to calicoblasts and underscore distinctions in gene expression between calicoblasts under both treatments. This indicates substantial changes in gene expression throughout development, suggesting mature calicoblasts may exhibit unique patterns and functions compared to newly developed calicoblasts.

Time-course analysis of single-cell level DEGs in bulk RNA-seq data

Given the differences observed in the single-cell transcriptomics, an intriguing question remains: Do these differences persist across the broader transcriptomic landscape and exhibit consistent patterns across various time points during regeneration? To address this knowledge gap, we focused on the transcriptome-wide scale, aiming to investigate whether the differences observed at the single-cell level are reflected in the bulk RNA-seq data. Upon examination, it appears that the dynamic expression changes of DEGs and biomineralization-related genes identified in the single-cell transcriptome do not exhibit significant patterns in bulk RNA-seq data (Fig. 6 and Supplementary Data 5). To gain deeper insights into the sustained effects of these genes during coral regeneration, we employed the TCseq method for clustering based on temporal gene expression patterns. The results indicate that genes significantly up- or downregulated in the single-cell transcriptomes do not uniformly exhibit similar changes in expression in the bulk RNA-seq data at the same time point (Fig. 6a, b).

For DEGs showing significant upregulation in the single-cell transcriptome, we categorized them into three clusters based on their temporal gene expression patterns (Fig. 6a and Supplementary Data 5). DEGs in cluster 1 exhibit significant downregulation in the early stages of coral biomineralization, followed by irregular fluctuations over time, with the majority peaking at day 21. Key members include genes possibly associated with biomineralization, such as *Pzp*, *calmodulin*, *Ca2*, and *calumenin*, along with stress-related genes like *Pxdn*. DEGs in cluster 2 display continuous fluctuation during coral skeletal reconstruction with an overall increasing trend until reaching relative stability in the later stages, including a group of alpha collagens. DEGs in cluster 3 show significant peaks in expression at the early

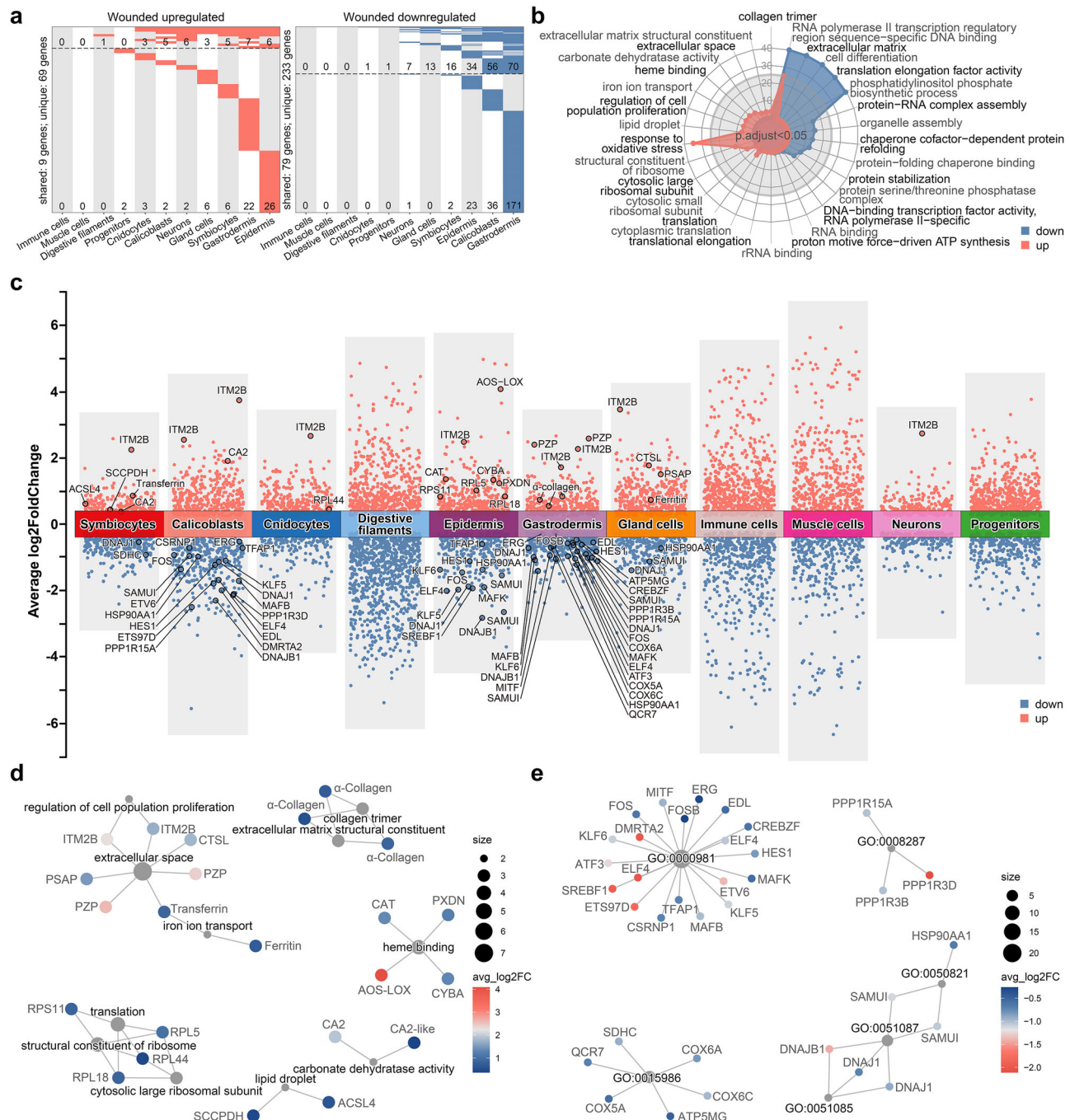


Fig. 3 | Analysis of differentially expressed genes in coral regeneration.

a Heatmaps illustrating the distribution of significantly differentially expressed genes (DEGs) between the wound and unwound treatments. The left heatmap shows DEGs upregulated in the wound treatment relative to the unwound treatment, while the right heatmap displays DEGs downregulated in the wound treatment. Numbers below the dotted lines indicate DEGs specific to only one cell type, and numbers above the dotted lines represent DEGs identified in at least two cell types. **b** Radar plot depicting GO enrichment pathways of significantly DEGs. Each radial axis represents a distinct GO term, and each concentric circle indicates different levels of significance, represented by $p.adjust$ values. The colored areas represent the significance of each GO term for different treatments. GO terms inside the gray circle are considered significantly

enriched ($p.adjust < 0.05$). **c** Manhattan plot for significantly DEGs. The x-axis represents cell types, while the y-axis shows the average expression level of genes in each cell type. Each point in the plot represents a gene. Orange-red points correspond to upregulated DEGs in the wound treatment relative to the unwound treatment, while blue points represent downregulated DEGs. Genes that are annotated in the plot are considered to be significantly differentially expressed, based on the criteria of $|avg_log2FC| > 0.25$ and an adjusted p -value (p_val_adj) < 0.1 . **d** Cnetplot showing the network linking significantly upregulated DEGs to enriched GO pathways. **e** Cnetplot displaying a partial network linking significantly downregulated DEGs to enriched GO pathways, with the full network shown in Supplementary Fig. 10.

(day 3) and mid (day 15) stages, encompassing stress-related genes (*Itm2b*, *AOS-LOX*, *Ctsl*, and *ferritin*) and symbiosis-related gene (*transferrin*).

For DEGs exhibiting downregulation in the single-cell transcriptome, we clustered them into four groups (Fig. 6b and Supplementary Data 5).

DEGs in cluster 1 display a noticeable peak at day 15, including stress-related genes (*soma ferritin* and *Hsp90aa1*) and biomineralization-related genes [*Mmp25*, *Timp1*, *proteoglycan 4 (Prg4)*, and *Bp10*]. DEGs in cluster 2 exhibit a similar expression pattern to the first type of temporal gene

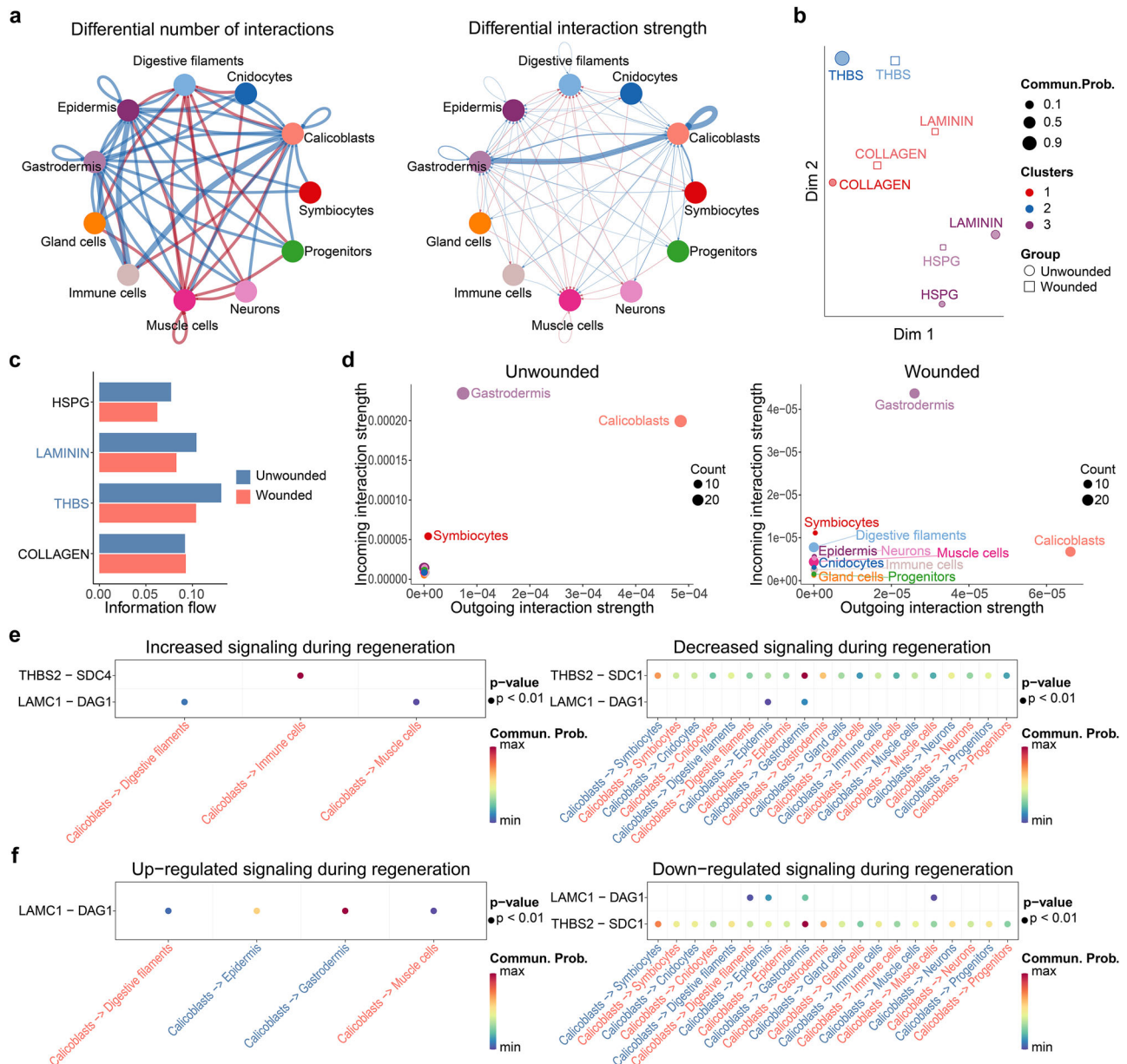


Fig. 4 | Cell-cell communication networks during coral regeneration. **a** Circle plot showing the differential number of interactions or interaction strength in the cell-cell communication network among different cell populations across two treatments. Edges colored in red (or blue) represent enhanced (or reduced) signaling in the wounded treatment compared to the unwounded treatment. **b** 2D visualization of signaling networks from the two treatments (wounded vs. unwounded) to identify signaling groups based on their functional similarity. The plot was generated using manifold learning and classification analysis performed by CellChat. Functional similarity was quantified by comparing the signaling pathways and ligand-receptor pairs across the two treatments, with similar senders and receivers clustered together. This analysis highlights potential alterations in the signaling pathways between the two conditions, with each point representing a signaling group and its relative position reflecting the degree of functional similarity. **c** The bar chart compares the overall information flow (interaction strength) of each signaling pathway between wounded and unwounded treatments, identifying pathways that (i) turn off, (ii) decrease, (iii) turn on, or (iv) increase based on changes in their information flow between the two conditions. The signaling pathways are ranked according to the differences in their overall information flow between the two treatments. A paired Wilcoxon test was performed to assess the significance of the differences in signaling information flow between the two conditions. Significantly altered pathways, with a

higher information flow in the unwounded treatment, are highlighted in blue. **d** 2D visualization showing outgoing and incoming interaction strengths across different cell types, highlighting significant changes in the sending or receiving signals between treatments. The x-axis represents outgoing interaction strength, and the y-axis represents incoming interaction strength, allowing identification of cell populations with altered signaling activity in response to the different treatments. **e** Changes in the communication probability of ligand-receptor pairs from calicoblasts to other cell types, showing increased and decreased signaling in the wounded treatment compared to the unwounded treatment. Increased signaling indicates a higher communication probability (strength) in the wounded treatment, while decreased signaling indicates a lower communication probability in the wounded treatment compared to the unwounded treatment. The red/blue color scale represents directional changes (increase/decrease), with full quantitative data provided in Supplementary Data 4. **f** Changes in the communication probability of ligand-receptor pairs from calicoblasts to other cell types, showing up- and downregulated signaling in the wounded treatment compared to the unwounded treatment. Differential expression analysis was performed to identify these changes based on the fold change of ligands in the sender cells and receptors in the receiver cells. Complete numerical details for these ligand-receptor pairs are available in Supplementary Data 4.

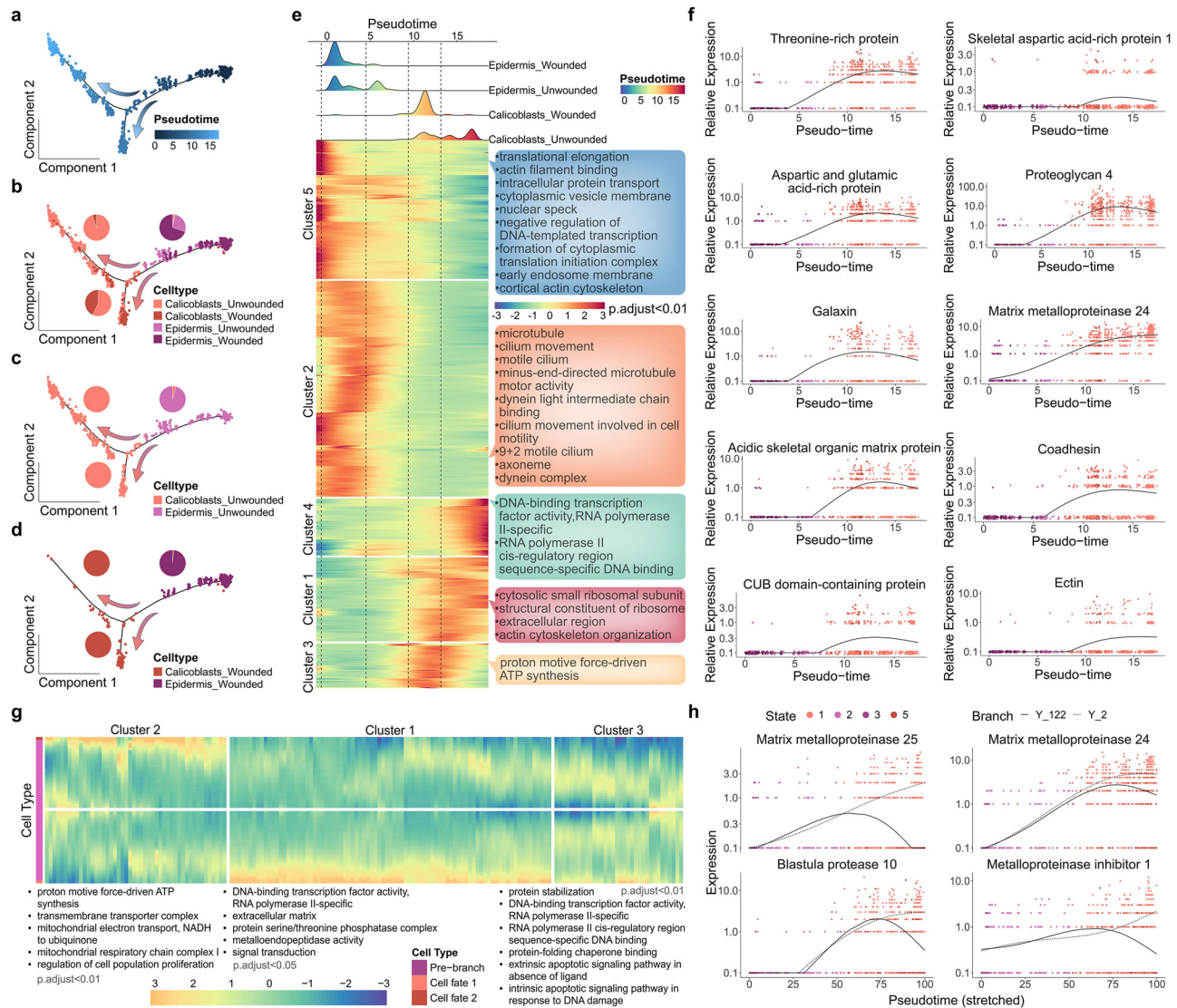


Fig. 5 | Pseudotime trajectory analysis during coral regeneration. **a** Pseudotime analysis of epidermal cells and calicoblasts combining both wounded and unwounded treatments, illustrating their developmental trajectories. Colors transition from dark to light, representing progression from early to late developmental stages along the inferred pseudotime axis. **b** Pseudotime trajectories of epidermal cells and calicoblasts from both treatments combined. Epidermal cells are shown in light purple (unwounded) and dark purple (wounded), while calicoblasts are represented by light orange (unwounded) and dark orange (wounded). **c** Pseudotime analysis of epidermal cells and calicoblasts in the unwounded treatment. Epidermal cells are displayed in light purple, and calicoblasts in light orange. **d** Pseudotime analysis of epidermal cells and calicoblasts in the wounded treatment. Epidermal cells are shown in dark purple, and calicoblasts in dark orange. **e** Ridge plot and heatmap visualization of pseudotime analysis. The upper panel shows a ridge plot illustrating the distribution of cell densities along the pseudotime axis for four groups: epidermal cells from the wounded treatment, epidermal cells from the unwounded treatment, calicoblasts from the wounded treatment, and calicoblasts from the unwounded treatment. The lower panel presents a heatmap of clustered gene expression profiles associated with intercellular state transitions along

pseudotime. GO functional enrichment analysis was performed for each gene cluster, with significantly enriched GO terms (p.adjust<0.01) labeled next to the corresponding clusters. **f** Gene expression dynamics of biom mineralization-related genes are visualized along the pseudotime trajectory. The x-axis represents pseudotime, indicating the progression of cellular development, while the y-axis shows the normalized expression levels of selected genes. Cells are colored by their assigned cell types, consistent with the color scheme used in panel b, highlighting distinct gene expression patterns associated with specific cell populations during developmental transitions. **g** Cell fate-specific heatmap and GO enrichment analysis. The upper heatmap displays gene expression patterns along different cell fates across the pseudotime trajectory. Cells are grouped by their assigned fate, highlighting distinct expression profiles associated with specific developmental outcomes. The lower section of the plot shows the GO enrichment analysis for each cluster. Only GO terms with a p.adjust < 0.01 are shown. **h** Gene expression dynamics of genes enriched in the GO terms "extracellular matrix" and "metallopeptidase activity" in Cluster 1 from panel g, displayed along the branched pseudotime trajectory. Cells are colored according to their assigned cell state.

expression pattern observed in the upregulated genes in the single-cell transcriptome, primarily involving a set of ribosomal proteins, ATP synthase-related genes, cytochrome c oxidase, and a biom mineralization-related gene (*Usomp4*). DEGs in cluster 3 show significant downregulation at days 15 and 21, while DEGs in cluster 4 exhibit significant downregulation in the early stages of coral skeletal reconstruction, followed by varying degrees of recovery at days 18 and 27, including *Mmp24* and *Timp4*.

Concerning biom mineralization-related genes, we categorized them into three classes (Fig. 6c and Supplementary Data 5). The results reveal that genes involved in calcification vary across different stages of coral regeneration. However, their significant fluctuations are more pronounced before the mid-stage of skeletal reconstruction, subsequently stabilizing in the later stages. As the biom mineralization progresses to later stages, the expression levels of these genes become more stabilized, indicating a potential shift in

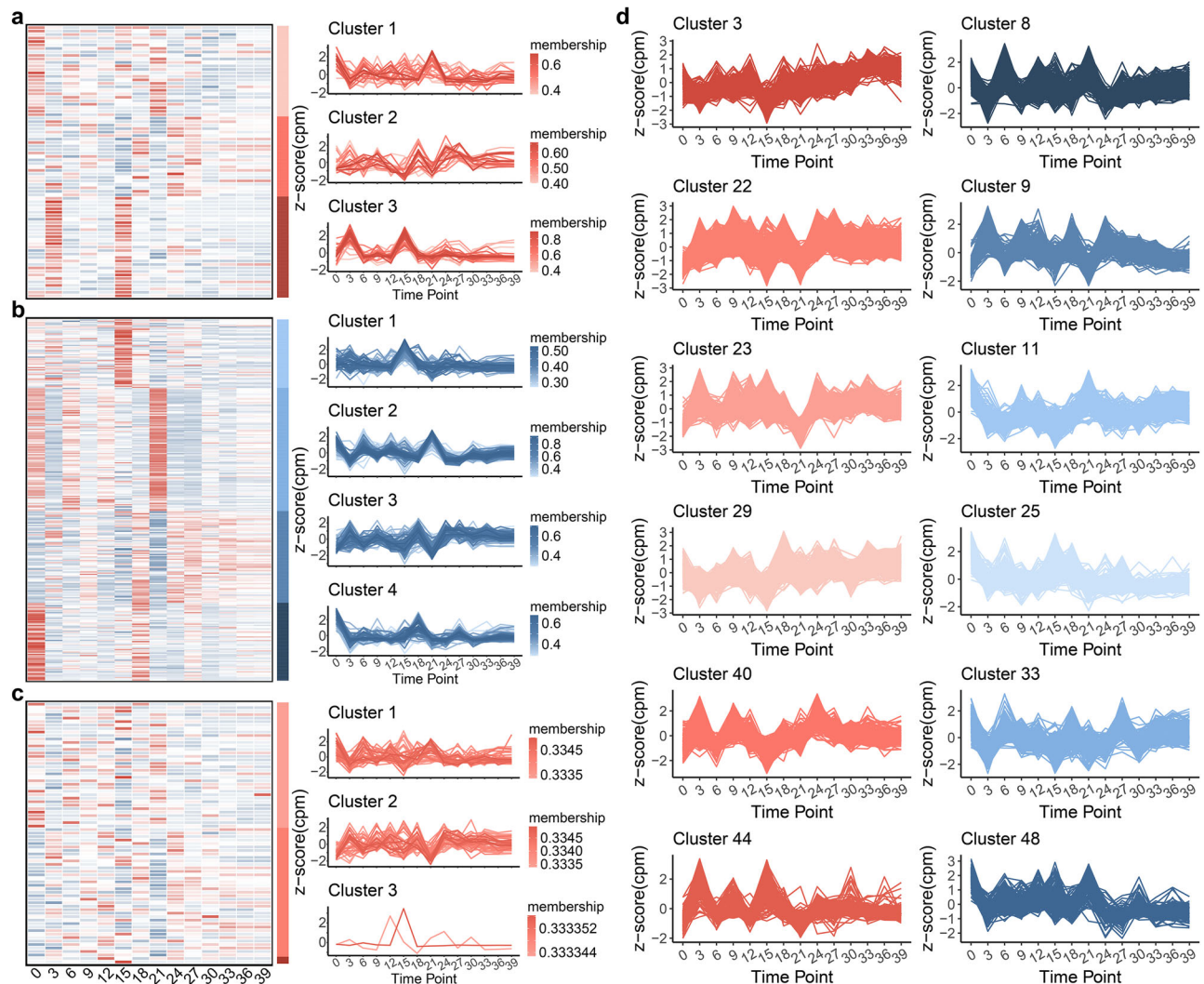


Fig. 6 | Temporal analysis of DEGs in bulk RNA-seq data. Temporal analysis of significantly upregulated (a), downregulated (b), and biomineralization-related (c) DEGs at the single-cell level in wounded treatment compared to unwounded treatment. On the left, heatmaps display gene expression across different clusters, with the x-axis representing days post-injury (with day 0 as the control, unwounded sample) and the y-axis representing clusters of genes. On the right, dynamic gene

expression profiles, generated using the TCseq method, show temporal trends for each cluster, with membership values indicating the degree to which genes belong to a particular cluster. **d** Time series dynamic gene expression profiles of DEGs, selected from the 12 colored profiles in Supplementary Fig. 19, representing clusters of genes with statistically significant expression patterns across time, visualized here using the TCseq package to highlight temporal gene expression dynamics.

focus or a reduced requirement for their active participation in the later phases of coral regeneration.

To further explore temporal gene expression patterns during coral skeletal reconstruction, STEM software identified 12 clusters with significant temporal features based on genes showing significant differential expression at least at one time point ($|\log_2\text{FoldChange}| > 1$ and $\text{padj} < 0.05$) (Fig. 6d, Supplementary Fig. 19, and Supplementary Data 5). Gene expression changes showed a noticeable reduction in magnitude in the later stages of the study. Among the USOMPs, clusters 3, 11, 22, 40, and 48 exhibited distinct expression patterns. Notably, *Usomp4* (in cluster 11) and one isoform of *Usomp5* (in cluster 48) displayed significant downregulation in the early reconstruction stages, with a partial recovery but consistently lower expression levels than the normal state throughout the study period. Similarly, *Usomp7* and *Usomp1* (both in cluster 11) showed significant downregulation on the third day post-injury, maintaining consistently lower expression levels thereafter. In contrast, only *Usomp3* (in cluster 3) and another isoform of *Usomp5* (in cluster 22) displayed upregulation during certain periods of coral skeletal reconstruction. *Galaxin 2* (in cluster 29) remained significantly upregulated in the middle to later stages. The expression of acidic skeletal organic matrix protein (in cluster 3) and *Saarp1*

(in cluster 22) fluctuated in the early stages but maintained relatively high expression levels in the later stages. Particularly, *Ca2* (in cluster 29) consistently and significantly increased in fluctuations, with high expression levels throughout the study.

In addition, we utilized PacBio sequencing technology to conduct comprehensive full-length transcriptome sequencing on coral samples collected on the 21st day post-injury, as well as from unwounded coral specimens. Our analysis did not reveal substantial distinctions in critical aspects such as alternative splicing, alternative polyadenylation, and fusion genes between these specified treatments. In tandem, an examination through miRNA sequencing failed to discern any noteworthy differences in this particular domain. In summary, these results provide an exploration of the transcriptomic landscape in coral samples undergoing regeneration. The investigation spans from single-cell transcriptomics to bulk RNA-seq data, encompassing various time points during the skeletal reconstruction process.

Discussion

The expeditious regeneration of coral colonies is essential for their survival, as demographic indicators, including growth, fecundity, and mortality,

exhibit a pronounced reliance on size⁴⁴, which implies a prioritization of energy allocation. Moreover, coral colonies demonstrate a more extensive level of physiological integration than previously recognized, coordinating energy and resources across the entire colony to support regeneration⁴⁵. Acute mechanical damage, caused by factors such as tropical storms, predation, and fragmentation for transplantation, can lead to increased mortality rates, and even loss of reproductive capacity⁴⁶. Previous research has shown a widespread decline in reproductive output in locations distal to the damaged tissue⁴⁵, suggesting that the regeneration from injury in corals may necessitate extended colony integration, challenging the notion that the energy requirements for this crucial process are solely provided by polyps directly adjacent to the damaged area. Therefore, facilitating the rapid and substantial growth of transplanted corals becomes imperative to ensure their survival and successful integration into the reef ecosystem.

Here, our morphological study identified a peak phase in the recovery of *A. muricata*, occurring approximately 2–4 weeks post-injury (Fig. 1). Building upon this observation, we established the *A. muricata* cell atlas using scRNA-seq to characterize the cellular dynamics and molecular processes underlying the regeneration phenomenon (Fig. 2). The analysis revealed that immune cells were abundant in both cnidocytes and digestive filament cells, suggesting potential immune-related gene or process involvement. Given cnidocytes' role in prey capture and anti-predator defense⁴⁷, and digestive filaments' function in food acquisition⁴⁸, these cell types may exhibit immune functions. Additionally, the findings demonstrated a consistent repertoire of cell types between unwounded and wounded samples, without the generation of novel cell subtypes. This observation may suggest that *A. muricata* primarily relies on the activation and functional adaptation of existing cell types to meet the demands of tissue repair and biomineralization, rather than forming new cell identities. Such a strategy may reflect an efficient regenerative approach, utilizing established cells for injury repair without invoking additional differentiation pathways. Nevertheless, discernible adjustments in the quantity of specific cell types during regeneration were noted, particularly an increase in epidermal cells and a decrease in calicoblasts, suggesting their specific involvement in the regeneration process as they adjust to meet functional and physiological demands.

Building on these findings, we further identified potential molecular markers by analyzing genes that were highly expressed and conserved across different treatments. Previous studies, such as Yoshioka et al.⁴⁹, have identified various symbiosis-related genes in *Acropora* corals. Our analysis focused on sulfate transporter, saccharopine dehydrogenase, and *Ca2* as notable markers, complementing broader efforts on symbiosis-related genes in corals. Sulfate transporters may play a critical role in sulfur metabolism during early coral life stages, supporting essential sulfur processes that facilitate symbiotic interactions⁴⁹. Previous studies have confirmed that the presence of symbiotic algae induces the upregulation of coral sulfate transporter homologs⁴⁹. Similarly, saccharopine dehydrogenase, involved in lysine degradation⁵⁰, was detected exclusively in symbiotic anemones and was absent in aposymbiotic ones⁵¹, suggesting it may enhance nitrogen transport and cycling within the symbiotic host. Additionally, *Ca2* is involved in ion transport and pH homeostasis, playing a crucial role in carbon-concentrating mechanisms essential for symbiont photosynthesis⁵². In cnidocytes, we identified high expression of leucine-rich repeat extensin-like proteins. This observation aligns with Levy et al. (2021), whose data also highlighted elevated expression of these proteins in this specific cell type³⁷. Together, these findings unveil a range of potential cell markers that contribute to our understanding of the functional roles and specialized characteristics of diverse cell types within coral organisms.

Previous research has highlighted genes containing a conserved BRI-CHOS domain (PF04089) as being differentially expressed in corals exposed to heat and acidification stress⁵³. In our study, we observed that *Itm2b*, which contains this domain, was specifically highly expressed in seven cell types under the wounded treatment. This suggests that *Itm2b* may play a critical role in the coral's response to environmental stress, particularly during regeneration. In addition, epidermal cells exhibited specific upregulation of a group of oxidoreductases during coral regeneration, with the most

prominent being AOS-LOX, a pathway present across all cnidarian lineages, which is capable of producing oxylipins, crucial stress mediators that indirectly regulate the expression of defense genes⁵⁴. Previous studies have indicated a significant upregulation of AOS-LOX in tissues near the wound and distal parts of coral colonies in the soft coral *Capnella imbricata* after injury, reaching peak expression at 1 h and 6 h post-injury, respectively⁵⁵. This suggests that the AOS-LOX pathway serves as a rapidly initiated stress response to acute mechanical damage. However, in our study, by integrating the temporal gene expression patterns of AOS-LOX (Supplementary Data 5), we observed significant up- and down-regulation at multiple time points during coral regeneration, particularly showing early-stage upregulation until later stages where expression stabilized. This implies that AOS-LOX may function as an enzyme with prolonged activity throughout the coral regeneration process. Because the activity of the enzyme can be influenced by various factors, despite gene expression is downregulated, previously synthesized enzyme molecules may persist and maintain activity for a certain period. Therefore, the intricate temporal regulation of AOS-LOX suggests its potential as a long-acting enzyme during coral regeneration. Several other oxidoreductases, including CAT, CYBA, and PXDN, also exhibited distinct regulation at certain time points during coral injury. CAT catalyzes the degradation of hydrogen peroxide, scavenging reactive oxygen species (ROS) and promoting cell growth, playing a crucial role in the coral response to external stimuli⁵⁶. CYBA, a subunit of NADPH oxidase (NOX), is involved in ROS production and plays an essential role in the immune system, with upregulation observed in response to coral fragmentation²³. PXDN, another key immune response gene, has been significantly upregulated in response to stony coral tissue loss disease⁵⁷ and heat stress⁵⁸. Therefore, forthcoming studies should incorporate assessments of enzyme activity to holistically comprehend the functional roles of these enzymes in response to coral wounding.

In addition to stress-related responses, metabolic processes also appear to be critical during regeneration. Notably, we identified *Acs4* as upregulated in symbiocytes under the wounded treatment. Previous studies suggest that *Acs4* provides essential components for energy storage and nutrient availability in corals, which could be vital for meeting the high energy demands of regeneration⁵⁹. Conversely, the downregulation of certain genes observed during coral regeneration suggests a strategic reallocation of cellular resources, prioritizing functions essential for tissue repair and recovery. Specifically, the suppression of genes related to protein synthesis and mitochondrial ATP production may indicate a temporary reduction in energy-intensive processes, such as protein translation and oxidative phosphorylation. This metabolic adjustment likely reflects the high energy demands of regeneration, as cells divert energy to critical processes like stress response and skeletal reconstruction. Similarly, the downregulation of genes involved in transcriptional regulation and protein folding implies a shift away from growth-related activities, redirecting resources to maintain cellular homeostasis and support recovery. These observations underscore the intricate balance of metabolic priorities during coral regeneration, ensuring efficient recovery while minimizing energy expenditure on non-essential functions.

Developmental trajectory analysis disclosed a continuous transformation of epidermal cells into calicoblasts during coral regeneration. Notably, epidermal cells predominantly transformed into a specific subtype of calicoblasts during this process. This observation suggests that the primary types of calicoblasts involved in biomineralization differ between normal coral growth and regeneration, reflecting the dynamic nature of coral cell differentiation and functional adaptation in response to distinct developmental and regenerative demands. The predominance of one calicoblast subtype during regeneration may indicate that this subtype represents a transitional state, tailored to the immediate requirements of the regenerative process. In later stages, this transitional calicoblast subtype could potentially revert to normal developmental patterns and transform into another subtype, as observed during typical coral growth. These findings imply that the two calicoblast subtypes observed in unwounded samples may not represent fundamentally distinct cell types but rather reflect

cells at different stages of development. During regeneration, both epidermal cells and calicoblasts in earlier developmental stages show a significant increase in numbers, suggesting an enhanced capacity to differentiate more epidermal cells into calicoblasts. This increase in early-stage calicoblasts may reflect distinct biomineralization potential compared to mature calicoblasts. Supporting this, in the early stages of calicoblast development, a noteworthy upregulation of the *Ca2* gene was observed, responsible for supplying essential carbonate ions for calcification⁵². Additionally, there was a significant augmentation in the secretion of skeletal organic matrix proteins, pivotal components of the coral skeleton⁶⁰. Moreover, we also identified other genes potentially involved in coral biomineralization that were specifically upregulated in cluster 1. Notably, PRG4 is a key protein, as existing studies suggest that the biomineralization process begins with the secretion of a proteoglycan matrix⁶¹, further confirming that this period corresponds to the early stages of calicoblast development. MMP24, belonging to the matrix metalloproteinase gene family, which plays a vital role in balancing and regulating the degradation of ECM proteins⁶², might also participate in coral calcification. In addition, distinct biomineralization-related genes were also found in other clusters. These findings underscore the potential variability of key genes responsible for biomineralization in calicoblasts at different developmental stages, and this variability might be intricately linked to the underlying mechanisms governing coral skeletal formation.

These findings are further supported by the results of the cell fate analysis, which revealed distinct gene expression patterns and functional enrichment across the two calicoblast fates identified. Cell fate 1, predominantly composed of calicoblasts from the unwounded treatment, was enriched in genes associated with extracellular matrix remodeling and stabilization. These genes play critical roles in creating and maintaining the extracellular matrix⁶², suggesting that calicoblasts in cell fate 1 may prioritize structural support and maintenance during normal coral growth. In contrast, cell fate 2, which includes the majority of calicoblasts from the wounded treatment and a portion from the unwounded treatment, exhibited enrichment in genes such as *Ca2*, *galaxin*, and *Mucin*, associated with biomineralization. These expression patterns suggest that calicoblasts in cell fate 2 may play a critical role in matrix deposition and calcification during regeneration. The gene enrichment in processes related to ATP synthesis, transmembrane transport, and cell proliferation suggests that these calicoblasts are energetically and metabolically prepared to support the demands of skeletal repair during regeneration. The distinct gene expression profiles observed in the two calicoblast fates further underscore the variability in cellular functions during coral development and regeneration. Importantly, these results align with the earlier hypothesis that the two calicoblast clusters may represent different stages of development rather than fundamentally distinct cell types. During regeneration, the observed predominance of calicoblasts associated with cell fate 2 could indicate a transitional state tailored to meet the immediate demands of skeletal repair, while calicoblasts associated with cell fate 1 might represent cells engaged in long-term skeletal maintenance. Together, these findings highlight the dynamic nature of coral calicoblasts and their ability to adapt their molecular functions in response to changing physiological needs. This flexibility ensures the effective fulfillment of distinct requirements during normal growth and regeneration, emphasizing the intricate coordination of developmental and regenerative processes in corals.

In our study, we observed that some of the upregulated, downregulated, and biomineralization-related DEGs in the wounded treatment of scRNA-seq displayed opposite expression patterns in the bulk RNA-seq data (Cluster 2 in Fig. 6a–c). This suggests that there is significant cellular heterogeneity in gene expression during the injury response, which is not fully captured by bulk RNA-seq. Since bulk RNA-seq represents an average gene expression across the entire sample⁶³, it may obscure the specific expression patterns of smaller cell populations, particularly those that are critical for regeneration but less abundant in the sample. In contrast, single-cell RNA-seq provides a more detailed view, capturing the molecular differences within individual cells, which is

essential for understanding complex biological processes like injury repair⁶⁴. This observation underscores the necessity of single-cell RNA-seq technology in studying intricate biological processes such as coral regeneration. By enabling the identification of gene expression variations at the single-cell level, single-cell RNA-seq allows for a more precise understanding of cellular dynamics and molecular mechanisms during the repair process, which might be missed when using bulk RNA-seq alone.

Although we observed some gene expression discrepancies between the single-cell and bulk RNA-seq data, the overall temporal gene expression analysis revealed that, after an initial period of dynamic gene activity, the expression patterns stabilize as the regeneration process progresses. This stabilization may signify a transition from the rapid regeneration phase to a state resembling normal growth, as the coral gradually returns to its steady-state physiological processes following the completion of tissue repair and biomineralization. Building upon this, we hypothesize that the enhanced biomineralization observed during the short-term coral repair stems from the generation of a substantial quantity of newly formed calicoblasts endowed with high biomineralization capability. This provides an explanation for the immediate short-term enhanced skeletal deposition observed in coral microfragments following initial fragmentation and tissue damage^{21,22}. However, the precise mechanisms governing the transformation of epidermal cells into calicoblasts remain elusive. Regarding the question of why corals accelerate the deposition of their skeletons in the early stages of regeneration, similar to other organisms, the coral skeleton plays a crucial role as an attachment substrate for the soft and vulnerable polyp. It acts as the primary physical barrier to protect the soft tissue from external environmental influences⁶⁵. Simultaneously, due to its richness in peptide neurotoxins, the coral skeleton provides biochemical protection⁶⁶. Therefore, the observed brief period of rapid growth post-injury serves as a defensive strategy, reinforcing damaged tissues.

In our investigation of the regeneration of *A. muricata*, we primarily focused on understanding the molecular mechanisms underlying rapid skeletal remodeling following coral injury, particularly the cellular and transcriptomic dynamics involved. However, our study did not address the tissue repair and regeneration processes, which are equally crucial in coral regeneration. Existing literature suggests that the first step in coral regeneration is wound healing, a rapid tissue repair process that prevents infection and further tissue loss after injury⁶⁷. Therefore, investigating the mechanisms of tissue repair and regeneration is of paramount importance for a more comprehensive understanding of coral regeneration. Future research could build on our findings by focusing on functional validation of key genes and cellular dynamics involved in both tissue and skeletal regeneration. Moreover, integrating multi-omics approaches, including proteomics and metabolomics, as well as exploring species-specific variations and environmental influences, will help advance the field of coral regeneration research.

Methods

Sample collection and regeneration experiment

In the aquatic environment of the Xisha Islands (latitude: 15°40′ – 17°10′N, longitude: 111° – 113°E), three coral colonies (*A. muricata*), each approximately 20*20*25 cm in size, were collected at a depth of approximately 5 to 10 meters. After collection, the corals were transferred to laboratory aquariums for a one-month acclimation period. Subsequently, three branches, each measuring 3–5 cm and exhibiting normal growth, were selected as biological replicates for the unwounded group in the regeneration time-course experiment. Additionally, the tips of five branches (each not exceeding 1.5 cm) were excised for the preparation of cell suspensions in the unwounded group for single-cell experiments, and one branch, measuring 3–5 cm and displaying normal growth, was chosen as the unwounded group for the HRCT experiment.

Simultaneously, 47 branches measuring 3–5 cm were transversely cut at their apical tips, removing 1 cm from the apical region, which was set

aside for subsequent experiments. Among these, 39 branches were randomly assigned to the regeneration time-series experiment. After coral injury, three branches were collected as biological replicates at each of the following time points: 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, and 39 days post-injury. Additionally, five branches were allocated to the single-cell experiment group, with tips (not exceeding 1.5 cm) sampled at 21 days post-injury. Finally, three branches were used for HRCT 3D reconstruction experiments, with samples collected at 14, 21, and 28 days post-injury. Sampling was consistently performed around midday to minimize diurnal variability.

The coral specimens were cultivated in a standard RedSea tank (redsea575, Red Sea Aquatics Ltd., London, UK) using the Berlin Method. This method relies on live rocks for biological filtration, where aerobic bacteria convert harmful ammonia into nitrites and nitrates, and anaerobic bacteria further process nitrates into nitrogen gas. The tank maintained a temperature of $26 \pm 1^\circ\text{C}$, salinity at 35 ppt, and was equipped with three coral lamps (AI®, Red Sea Aquatics Ltd., London, UK), a protein skimmer (Reef Octopus Regal 250-S, Honya Co. Ltd., Shenzhen, China), a water chiller (tk1000, TECO Ltd., Port Louis, Mauritius), two propeller pumps (VorTech™ MP40, EcoTech Marine Ltd., Bethlehem, PA, USA) and a calcium reactor (Reef Octopus OCTO CalReact 200, Honya Co. Ltd., Shenzhen, China). The calcium reactor, containing coral skeleton fragments as the primary calcium source, maintained the calcium concentration between 350–450 ppm and alkalinity at 8–10 dKH. Corals were fed every other day with RedSea Reef Energy Plus Coral Nutrition AB+. To maintain optimal seawater parameters, magnesium levels were controlled between 1200–1350 ppm, potassium between 380–420 ppm, nitrate below 10 ppm, nitrite at 0, and phosphate under 0.02 ppm. The light intensity was maintained at approximately $400\text{--}500\ \mu\text{mol m}^{-2}\text{ s}^{-1}$ during the light period, with a light/dark cycle of 10 h of light and 14 h of darkness, simulating the natural photoperiod in the coral's native environment.

HRCT and 3D reconstruction

We utilized a 230-kV x-ray microfocus computed tomography system (Phoenix v|tome|x m, General Electric) at Yinghua NDT in Shanghai, China, to examine four *A. muricata* samples. Proprietary software from General Electric (GE) was employed to compile two-dimensional image reconstructions for each specimen based on matrices of scan slices. The relevant scan parameters for each sample were as follows: for the unwounded sample, voltage 120 kV, current 115 μA , voxel size 12 μm , timing 500 ms, number of images 2400, image width 1980 pixels, and image height 2000 pixels; for the 14-day post-injury sample, voltage 130 kV, current 60 μA , voxel size 6 μm , timing 500 ms, number of images 1,500, image width 2800 pixels, and image height 4000 pixels; for the 21-day post-injury sample, voltage 130 kV, current 100 μA , voxel size 9 μm , timing 500 ms, number of images 2500, image width 1,985 pixels, and image height 2,000 pixels; and for the 28-day post-injury sample, voltage 160 kV, current 70 μA , voxel size 9 μm , timing 500 ms, number of images 1600, image width 1500 pixels, and image height 4000 pixels. Image sections and 3D renderings were generated using VG Studio Max (v3.3.0).

Single-cell suspension preparation, library construction, and sequencing

To prepare the single-cell suspension, a 0.035 M PBS solution with a pH of 8.0 was created by diluting 0.02 M PBS (Solarbio, P1010-2L, Beijing, China). The samples were thoroughly washed with this buffered solution, a process repeated three times. A digestion solution was then prepared by diluting 1% Trypsin (Sigma-Aldrich, T1426-500MG, St. Louis, MO, USA) in the 0.035 M PBS solution. The obtained samples underwent a 2–3 h digestion at room temperature under sterile conditions, with the entire procedure performed in a clean bench. Subsequently, the solution was filtered using 40 μm Fisherbrand™ Sterile Cell Strainers (Thermo Fisher Scientific, 22-363-547, Waltham, MA, USA) and centrifuged at 350 g for 5–6 min. The resulting supernatant was collected, and the digestion was terminated with 0.4 mg

BSA (Sangon Biotech, A600332-0005, Shanghai, China) dissolved in 0.035 M PBS solution. Live cell counting and assessment of sample concentration and viability were performed using an automated cell counter (TC20™, Bio-Rad Laboratories, Inc., Hercules, CA, USA) with 0.4% trypan blue (Invitrogen, T10282, Carlsbad, CA, USA) at a 1:1 ratio to the cell suspension. The cells, considered qualified based on a viability greater than 60% and a concentration between 1×10^5 and 1×10^7 cells/mL, after washing and resuspension, were diluted to a concentration of 800 – 1200 cells/ μL to ensure optimal single-cell capture and then sent to Tianjin Novogene Bioinformatic Technology Co., Ltd. (Tianjin, China). Tianjin Novogene Bioinformatics Technology Co. carried out library construction using Chromium Controller (10x Genomics, Inc., Pleasanton, CA, USA), followed by quality control. Finally, sequencing was conducted on the HiSeq PE150 platform (Illumina, Inc., San Diego, CA, USA).

mRNA extraction, library construction, and sequencing

For the regeneration time-course experiment, we designed 14 time points, including the control state (0 days) and post-injury at days 3, 6, 9, 12, 15, 18, 21, 24, 27, 30, 33, 36, and 39. Each time point had three biological replicate samples, requiring a total of 42 Illumina cDNA libraries to be constructed (with mRNA > 1.5 μg for each sample). Additionally, at the control state and the highly active regeneration period (day 21 post-injury), one PacBio cDNA library (with mRNA > 10 μg for each sample) and three miRNA libraries (with mRNA > 2 μg for each sample) were required to ensure sufficient material for sequencing at these time points. For the construction of the PacBio cDNA library, an equal amount of mRNA from the corresponding three biological replicate samples was mixed together, totaling more than 10 μg .

During the mRNA extraction process, the mortar and pestle were consistently kept in liquid nitrogen to maintain sample integrity, strictly adhering to the manufacturer's instructions. The mRNA extraction steps were as follows: (1) Grinding coral samples into granules; (2) Adding TRIzol® LS Reagent (Thermo Fisher Scientific, 10296028, Waltham, MA, USA) in a ratio of approximately 1:3 (sample: reagent) to the ground samples, continuing grinding to powders, and allowing natural freezing; (3) Continuing to add a small amount of TRIzol® LS Reagent and grinding until the sample completely dissolved, then transferring the liquid to a 50 mL centrifuge tube; (4) Centrifuging at 4°C and 3150 g for 5–15 min, collecting the supernatant in a 50 mL centrifuge tube; (5) Adding BCP reagent (Molecular Research Center, BP 151, Cincinnati, OH, USA) in a ratio of 5:1 (sample: reagent) to the aforementioned centrifuge tube, shaking well, and letting it stand for 10 min; (6) Centrifuging at 4°C and 38,587.5 g for 15 min, collecting the supernatant, adding an equal volume of Isopropanol (Amresco, 0918-500 ML, Radnor, PA, USA), mixing well, and incubating overnight at -20°C ; (7) Centrifuging at 4°C and 38,587.5 g for 30 min, discarding the supernatant, and washing twice with diluted 75% Ice Ethyl alcohol, Pure (Sigma-Aldrich, E7023-500ML, Taufkirchen, München, Germany) to obtain the final mRNA product.

The samples were then sent to Tianjin Novogene Bioinformatic Technology Co., Ltd., for sequencing. The PacBio Sequel II platform was used for full-length transcriptome sequencing, Illumina HiSeq X Ten platform was used for short-read transcriptome sequencing, and the Illumina SE50 platform was used for miRNA sequencing. The company conducted quality control processing on the raw sequencing data.

Reference genome alignment

The single-cell transcriptome sequencing data were aligned to the reference genome assembly GCF_036669905.1 (NCBI RefSeq; BioProject: PRJNA544778) using Cell Ranger v6.0.0⁶⁸ software from 10x Genomics. Subsequently, gene expression quantification was performed to generate the cell-gene expression matrix. The clean reads from full-length transcriptome sequencing were mapped to the reference genome using GMAP v2017-06-20 software⁶⁹. For the processed small RNA reads with lengths ranging from 18 to 35 nucleotides, mapping to the reference genome and analysis were conducted using Bowtie v0.12.9⁷⁰. The reads mapped to the reference

sequences were aligned to the miRBase database v22.0 using mirDeep2 v2.0.0.5⁷¹ and srna-tools-cli to detect known mature miRNAs, and their expression levels and TPM values were quantified. Additionally, novel miRNAs were predicted through the integration of miREvo v1.1⁷² and mirDeep2.

The clean reads from short-read transcriptome sequencing were aligned to the reference genome using HISAT v2.0.4⁷³. The Cufflinks v2.1.1 Reference Annotation Based Transcript (RABT) assembly method⁷⁴ was employed to construct and identify both known and novel transcripts based on the HISAT alignment results. HTSeq v0.6.1 (-m union) was used for counting the read counts mapped to each gene⁷⁵. Subsequently, the mean transcripts per million (TPM) value for each gene across the three biological replicates was calculated, allowing for comparisons of gene expression levels between treatments and normalization for differences in library size.

Single-cell transcriptome analysis

The analysis and visualization of the cell-gene expression matrix were conducted using Seurat v4.9.9050 package⁷⁶ in R v4.3.1 software, with the following thresholds: cells containing between 200 and 2500 unique genes were retained, and genes expressed in at least 3 cells were included. To validate the reliability of cell clustering, we applied the UCell v2.8.0 package to score Seurat objects, while the resulting cell type composition within clusters was visualized using a customized plotting approach, inspired by the plot_attr function from the singleCellNet v0.1.0 package⁷⁷. The cell type distribution plots across different treatments utilized the dittoBarPlot function from the dittoSeq v1.13.1 package⁷⁸. The FindMarkers function in the Seurat package was employed for differential expression analysis between two treatments, considering genes with $|\text{avg_log2FC}| > 0.25$ and adjusted p-value ($p_{\text{val_adj}}$) < 0.1 as significantly DEGs. The clusterProfiler v4.8.3 package⁷⁹ was used for GO enrichment analysis of significantly DEGs. Cell-cell communication analysis for the two treatments was performed using the CellChat v1.6.1 package⁸⁰. Trajectory analysis for epidermal cells and calicoblasts was conducted using the monocle v2.28.0 package⁸¹.

For functional annotation, gene sequences were aligned to the NCBI non-redundant (nr) database using Diamond v0.9.14.115 with parameters (diamond blastx --db nr --query input.fasta --out output.xml --outfmt 5 --more-sensitive --max-target-seqs 20 --evaluate 1e-5)⁸². Protein functions were predicted using InterProScan v5.61-93.0 (/interproscan.sh -dp -ipr-lookup -goterms -pa -t p -i orfs.fasta -f xml) by analyzing protein structure domains and functional sites⁸³. Blast2GO v6.0.3 integrated the Diamond and InterProScan result files for final nr and GO annotation results⁸⁴. KEGG annotation was performed using KofamScan 1.3.0 (/exec_annotation -o kegg.out --cpu 8 --format mapper -e 1e-5 orfs.fasta)⁸⁵.

Other sequencing data analysis

The clustering analysis of short-read sequencing data in the regeneration time-course experiment was performed using the TCseq v1.22.6 package⁸⁶ and STEM v1.3.13 software⁸⁷. The input matrix for STEM analysis consisted of the temporal expression matrix of significant DEGs at least at one time point, derived from DESeq2 v1.38.3⁸⁸. Specifically, genes were considered significantly differentially expressed if they met the criteria $|\log_2\text{Fold-Change}| > 1$ and $\text{padj} < 0.05$. The union of these DEGs across all time points was used for the subsequent analysis. For full-length transcriptome, alternative splicing analysis utilized SUPPA v2.3 software⁸⁹, while alternative polyadenylation analysis employed TAPIS v1.2.1 software⁹⁰. Fusion gene identification criteria included alignment to two or more genomic loci in the reference genome (each aligned locus must be separated by at least 100 kb), with a minimum of two supporting reads in the corresponding short-read data. Differential expression analysis of miRNA utilized the DESeq2 package, considering genes with $\text{padj} < 0.05$ and $|\log_2(\text{fold change})| > 1$ as significantly differentially expressed. Additionally, the prediction of miRNA target genes was carried out using the intersection of the miRanda v3.3a⁹¹ and RNAhybrid v2.0⁹² software.

Statistics and reproducibility

Differential gene expression was quantified using the Seurat pipeline with the FindMarkers function, which identifies differentially expressed genes for each cell type between two treatments using the Wilcoxon Rank Sum test. Confidence intervals, sample sizes, and the number of replicates are reported where relevant. The sample sizes were determined based on previous studies and pilot experiments to ensure sufficient statistical power for detecting significant differences.

Ethics

All authors contributed to the manuscript preparation. This study complies with ethical guidelines. The coral samples were collected under permits issued by our collaborating institution, Hainan Academy of Ocean and Fisheries Sciences, in compliance with local and national regulations governing marine research. All ethical regulations relevant to environmental and marine scientific research were followed.

Reporting summary

Further information on research design is available in the Nature Portfolio Reporting Summary linked to this article.

Data availability

The scRNA-seq datasets generated in this study have been deposited in Figshare at <https://doi.org/10.6084/m9.figshare.25107968> and the National Center for Biotechnology Information (NCBI) under BioSample accession numbers SAMN39649577-80. The time-course short-read RNA-Seq data, Pacbio full-length RNA-Seq data, and miRNA-seq data are available in the NCBI under BioSample accession numbers SAMN39649527-68, SAMN16675175-76, and SAMN39649571-76, respectively.

Code availability

The R-code file can be found in the online repository on GitHub (https://github.com/tyhan2360/Coral_wounded).

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Author contributions

T.H. and C.H. performed the experimental studies. T.H. carried out the computational analysis. T.H. participated in the writing of the paper. Y.L. provided the sample resources. J.C. and Z.L. supervised the work. T.H., C.H., Y.L., H.Z., J.C., and Z.L. contributed to data discussion. All authors have read and approved the manuscript for publication.

Competing interests

The authors declare no competing interests.

Additional information

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